Ru Loaded CeO_2 Nanorods for Photothermal CO_2 Hydrogenation

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Objective and contribution

This project examined how support composition affects methane response over Ru-loaded CeO_2 nanorods under dilute CO_2 conditions. With Ru loading held at a common nominal target, CeO_2 modification provided a way to screen support formulations. I carried out support synthesis and modification, Ru deposition, reaction tests, gas chromatography (GC), and data interpretation, using the results to set subsequent composition conditions within the laboratory study.

Catalyst synthesis and screening design

CeO_2 nanorods were prepared by alkaline hydrothermal synthesis, with each modifier introduced alongside the Ce precursor. After calcination, nominal 2 wt% Ru was deposited by $NaBH_4$ reduction. Seven modifiers were screened against unmodified nanorods and P25 references before focused Zr-content experiments. Photothermal tests used approximately 130 mg of each catalyst with 0.5% CO_2 and 2.5% H_2 in Ar. The CH_4 GC peak area after 1 h served as a preliminary relative screening indicator.

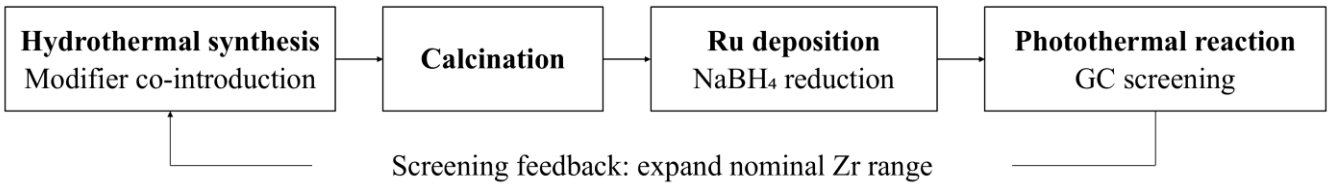


Figure 1. Catalyst preparation and composition screening. GC results guided subsequent nominal Zr-content ranges.

Support screening and Zr selection

Zr-modified CeO_2 gave the highest response (224.4 a.u.), approximately 20% above unmodified CeO_2 nanorods (187.1 a.u.) within the screening batch. This was the only tested modifier to exceed the unmodified reference, supporting a focused Zr-content study.

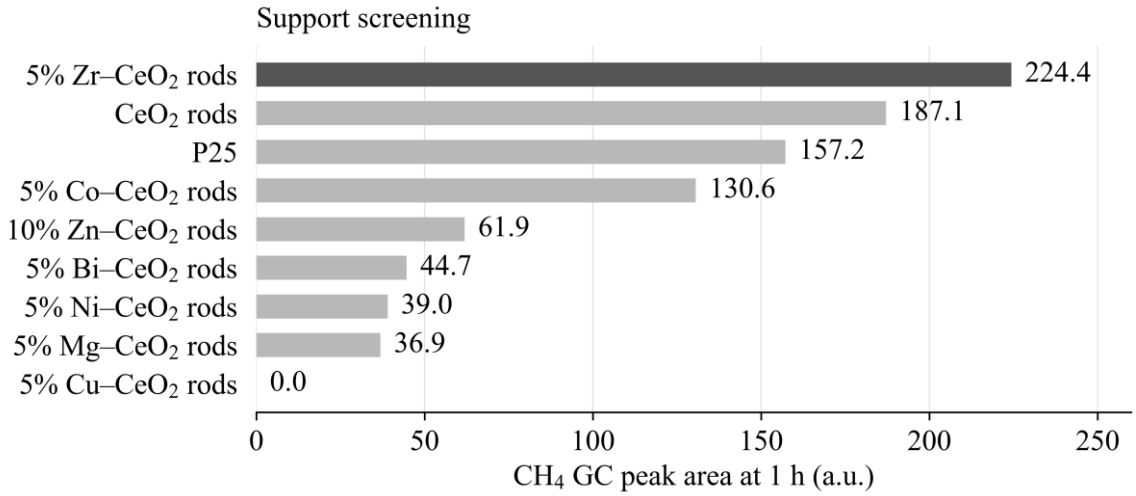


Figure 2. Support screening at nominal 2 wt% Ru. CH_4 GC peak areas after 1 h; dark gray identifies Zr-modified CeO_2. References are unmodified CeO_2 and P25. Nominal modifier contents: 10 mol% Zn; 5 mol% for the others.

Zr composition effects and interpretation

The July 0–10 mol% nominal Zr series peaked at 10 mol% (366.8 a.u.), supporting a wider range. The 5 mol% sample was lower than the 2.5 mol% sample; in August, 5 mol% was also lower than 1 mol%. Thus, increasing nominal Zr content did not consistently increase the response. In August, responses increased across 10, 20, and 30 mol%, with 30 mol% giving the highest tested response (268.7 a.u.).

The unmodified reference gave 187.1 a.u. in July and 161.3 a.u. in August, so raw peak areas were interpreted within each batch.

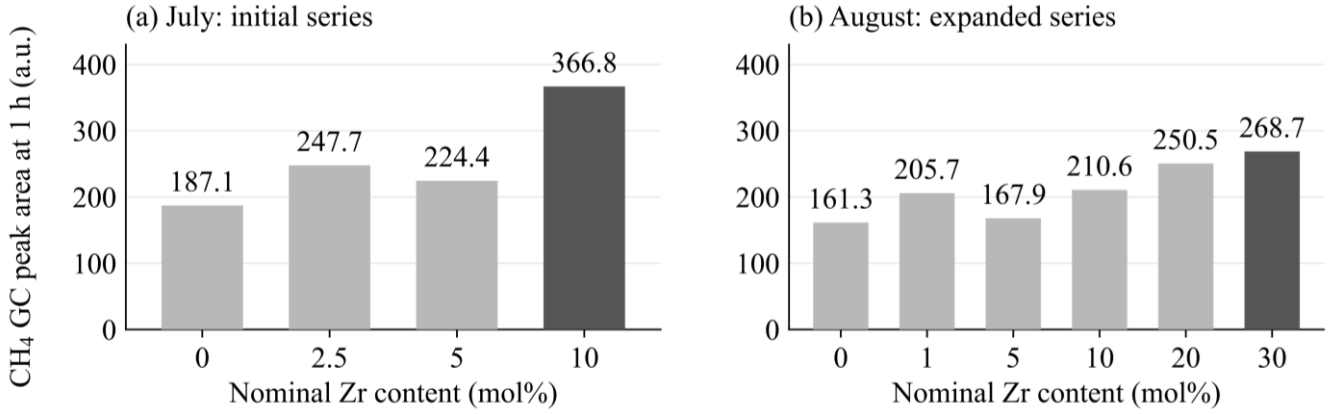


Figure 3. CH_4 GC response across nominal Zr contents (mol%). (a) July and (b) August batches, each measured after 1 h. Nominal 0 mol% denotes unmodified CeO_2. Raw peak areas are compared within each batch; compositions are nominal.

Both series had their highest responses at the upper tested boundary. These results supported further Zr screening without resolving an optimum.

Findings and scope

The study supports further investigation of Zr-modified CeO_2, with performance interpreted as within-batch GC response. Calibration and independent synthesis and test replicates are needed to establish quantitative performance. Elemental analysis is needed to confirm actual Ru and Zr contents relative to the nominal preparation compositions. Structural and surface characterization would test whether support defects or Ru-support interactions explain the observed response.